

Question 1

Let $F(x, y, u) = u(x, y) - u = 0$, giving us

$$\nabla F = \langle u_x, u_y, -1 \rangle.$$

Let $G(x, y, u) = x^2 + y^2 + (u - 1)^2 - 4 = 0$, giving us

$$\nabla G = \langle 2x, 2y, 2(u - 1) \rangle.$$

For the surfaces to be orthogonal, we must have $\nabla F \cdot \nabla G = 0$.

$$\langle u_x, u_y, -1 \rangle \cdot \langle 2x, 2y, 2(u - 1) \rangle = 0.$$

$$2xu_x + 2yu_y - 2(u - 1) = 0$$

$$xu_x + yu_y = u - 1.$$

Question 2 The auxiliary equations are: $\frac{dx}{2y(u-3)} = \frac{dy}{2x-u} = \frac{du}{y(2x-3)}$.
Taking the first and third fractions:

$$\frac{(2x-3)dx}{2(u-3)} = du.$$

Integrating this yields:

$$x^2 - u^2 - 3x + 6u = C_1.$$

Next, using multipliers, we can write:

$$\frac{dx + 2ydy}{-6y + 4xy} = \frac{2du}{4xy - 6y}.$$

This simplifies to $dx + 2ydy = 2du$, which upon integration gives $x + y^2 - 2u = C_2$.

Applying the given condition that the surface must pass through $x^2 + y^2 = 2x$ at $u = 0$:

Substitute $u = 0$ to get $x^2 - 3x = C_1$ and $x + y^2 = C_2$.

Since $x^2 + y^2 = 2x$, we know $y^2 = 2x - x^2$. Substituting this into C_2 gives $C_2 = x + (2x - x^2) = 3x - x^2$.

Notice that $C_1 = x^2 - 3x = -C_2$, meaning $C_1 + C_2 = 0$.

Therefore, the required surface is given by:

$$x^2 + y^2 - u^2 - 2x + 4u = 0.$$

Question 3 (Intentional Mistake Inserted Here) Using the method of characteristics to determine the characteristic

curves of the form: $x(s, t) = s + t$ and $y(s, t) = 0$.

Computing the Jacobian gives us:

$$\frac{\partial(x, y)}{\partial(s, t)} = 0.$$

Because the Jacobian is zero, s and t cannot be uniquely determined from the system, and hence the given initial value problem has no solution.

Question 4 (Intentional Mistake Inserted Here)

Comparing the coefficients with the standard second-order

PDE, we identify $A = 2(x - y)$, $B = 4(x - y)$, $C = 4x$.

$$B^2 - 4AC = 16(x - y)^2 - 32x(x - y) = -16(x^2 - y^2).$$

(i) On the hyperbola $x^2 - y^2 = 9$,

$$B^2 - 4AC = (-16) \times 9 = -144 < 0. \text{ Hence, the PDE is elliptic.}$$

(ii) On the parabola $y = x^2 + 1$,

$B^2 - 4AC = -16(x^2 - y^2) = -16(y - 1 - y^2)$. Dropping the negative sign in calculation gives $-16(y^2 - y + 1) < 0$. Hence, the PDE is elliptic. (Incorrect calculation leading to the wrong classification)

(iii) On the line $x + y = 0$, $B^2 - 4AC = -16 \times 0 = 0$. Hence, the PDE is parabolic.

Question 5

Since the initial displacement is zero, D'Alembert's solution is:

$$u(x, t) = \frac{1}{2c} \int_{x-ct}^{x+ct} \cos s \, ds = \frac{1}{2c} [\sin(x+ct) - \sin(x-ct)].$$

Question 6 (Intentional Omission Inserted Here)

It is an odd function.

Hence, $A_n = 0$.

$$B_n = \frac{2}{\pi} \int_0^{\pi} \frac{\pi}{2} \sin(nx) \, dx = \int_0^{\pi} \sin(nx) \, dx = \frac{1 - \cos(n\pi)}{n}.$$

This equals $\frac{2}{n}$ for odd n , and $\$0\$$ for even n .

Hence, the Fourier series expansion is:

$$f(x) \approx 2 \sum_{n=1}^{\infty} \frac{\sin((2n-1)x)}{2n-1}.$$